Singapore School Kinderland

Primary 3 IT • Chapter 1A: Parts of a Computer Hardware

OFFICIAL ANSWER SHEET (Pages 2 to 7)

1A.1 Introduction — Hardware vs Software — Worksheet Page 2

WORK TO DO – Hardware vs Software

1. Keyboard → **Hardware**
2. MS Paint program → **Software**
3. Monitor → **Hardware**
4. A computer game app → **Software**

1A.2 Computer Hardware & 1A.3a Input Devices — Worksheet Page 3

WORK TO DO – Sort the Hardware

External Hardware	Internal Hardware
• Monitor • Keyboard • Printer	• Motherboard • Memory Card (RAM) • CPU Chip

WORK TO DO – Match the Input Device

Input Device	Correct Function (Match)
Light Pen	→ **Has a light-sensitive detector at its tip**
Barcode Reader	→ **Reads barcode using a beam of light**
Trackball	→ **A ball you roll with fingers or palm**
Touch Pad	→ **Senses finger movements on a pad**
Joystick	→ **Moves with a lever in all directions**

1A.3b Output Devices & 1A.3c Central Processing Unit (CPU) — Worksheet Page 4

WORK TO DO – Printer Types

1. Uses an ink ribbon and striking pins to make dots → **Dot Matrix Printer**
2. Sprays ink on paper for high quality text & graphics → **Inkjet Printer**
3. Uses laser beams; fast but expensive → **Laser Printer**

WORK TO DO – Who Does What? (CU or ALU)

1. Controls all actions of a computer → **CU (Control Unit)**
2. Adds two numbers together → **ALU (Arithmetic Logic Unit)**
3. Compares if A is greater than B → **ALU (Arithmetic Logic Unit)**

1A.3d Ports & 1A.3e Modem

Worksheet Page 5

WORK TO DO – True or False

1. [**T**] A Serial Port allows only 1-bit transmission at a time.
2. [**T**] USB stands for Universal Serial Bus.
3. [**F**] A Parallel Port cannot connect a printer. *(False – Parallel port connects printers)*

WORK TO DO – Fill in the Blanks

1. Modem stands for Modulator - **Demodulator**.
2. A modem that stays inside the computer is called a/an **Internal** modem.

1A.3f-g Motherboard & Disk Drive • Chapter 1A Review

Worksheet Page 6

WORK TO DO – Label & Define

1. What is the function of a motherboard?

Answer: The motherboard is an internal device that allocates power and allows communication between the CPU, memory (RAM), disk drives, and other input/output devices.

2. What is the difference between a Track and a Sector on a disk?

Answer: A **Track** is a concentric circular ring on the disk surface, while a **Sector** is a smaller pie-shaped segment/division of a track.

⭐ CHAPTER 1A — REVIEW EXERCISE

Q1. Fill in the blanks:

i. **CPU** is known as the brain of the computer.
ii. Task of a **Modem** is to convert signals over the Internet.
iii. Physical movement of a mouse is displayed on the **Monitor**.
iv. **Hardware** are the physical components that a computer is made up of.

Q2. Write the full form of the following:

i. **CPU** → **Central Processing Unit**
ii. **USB** → **Universal Serial Bus**
iii. **LCD** → **Liquid Crystal Display**
iv. **RAM** → **Random Access Memory**

Q3. Write the difference between:

i. Scanner and Printer:

• **Scanner:** An *input device* that captures physical documents/images and converts them into digital files.
• **Printer:** An *output device* that transfers digital text/images from a computer onto paper.

Q3. Write the difference between (Continued):

ii. Mouse and Trackball:

• **Mouse:** The entire device is moved across a surface to move the cursor.

• **Trackball:** The device remains still; only the ball on top is rolled with fingers to move the cursor.

iii. Internal and External Hardware:

• **Internal Hardware:** Located inside the computer casing (e.g., Motherboard, RAM, CPU chip).

• **External Hardware:** Connected to the computer from the outside (e.g., Monitor, Keyboard, Printer).

iv. Hardware and Software:

• **Hardware:** Physical components of a computer that you can see and touch (e.g., Keyboard, CPU).

• **Software:** Programs and apps running on hardware that tell the computer what to do (e.g., MS Paint).

Q4. Rewrite the sentences after correcting them:

i. *Tracks are triangular rings on a disk.*

→ **Corrected:** Tracks are **circular** rings on a disk.

ii. *Motherboard is an external hardware device.*

→ **Corrected:** Motherboard is an **internal** hardware device.

iii. *All actions of a computer are controlled by Arithmetic and Logic Unit.*

→ **Corrected:** All actions of a computer are controlled by the **Control Unit (CU)**.

iv. *A scanner is an output device.*

→ **Corrected:** A scanner is an **input** device.

Q5. Write on:

i. Motherboard:

The motherboard is the primary internal printed circuit board (PCB) of a computer. It holds the CPU, memory slots, and expansion cards, allocates power from the power supply, and enables high-speed communication between all internal and external devices.

ii. Keyboard:

A keyboard is the most common input device. It features alphanumeric, numeric, navigation, and function keys that allow the user to input text, numbers, symbols, and shortcut commands into the computer.

iii. Touch Screen:

A Touch Screen is a display screen that also acts as an input device. It senses the touch of a user's finger or a stylus pen to select icons, type, and give commands directly on the screen without needing a separate mouse or keyboard.

iv. CPU (Central Processing Unit):

The CPU is known as the "Brain of the computer." It carries out and processes all the instructions, calculations, and commands given to the computer. It consists of two main internal parts: the **Control Unit (CU)**, which directs all computer operations, and the **Arithmetic Logic Unit (ALU)**, which performs all math and logic operations.